

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8500**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600028
Course Section:	1				

L_29_485_986_L_1505_38411



Name of the candidate. Shrihari Mahabal

Candidate's UID number:

2 0 2 2 6 0 0 0 2 8

Examination class room number:

4 0 1

Invigilator's signature with date:

8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH-MSE

(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: CSE-AI ML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 08/10/2025

COURSE NAME: Deep Learning (DL)

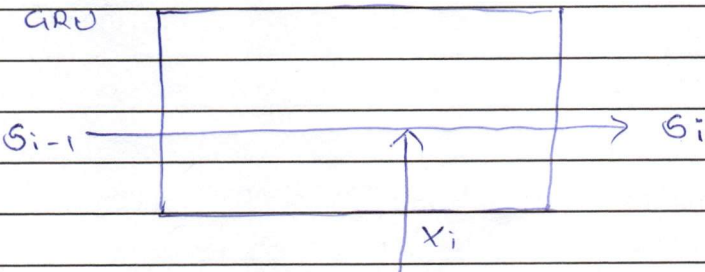
(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left-hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	3b)	The architecture of GRU (Gated Recurrent Unit) is shown below.  It solves the problem of RNN which tends to forget the previous inputs through usage of input & output gates (forget & update gates) to enable selective selection or rejection of information. Input Gate: $i_t = \sigma(W_i s_{t-1} + U_i x_t + b_i)$ Output Gate: $o_t = \sigma(W_o s_{t-1} + U_o x_t + b_o)$ Here, W_i, U_i, U_o, W_o are trainable (learned) weight matrices The information given through input is given by $\tilde{s}_t = \sigma(o_t \cdot (W s_{t-1} + U x_t + b))$ here, o_t is the output filter which decides what information is to be written. The state s_t is given by $s_t = (1 - i_t) s_{t-1} + i_t \tilde{s}_t$ Here, i_t (input gate) filters the information we get from input x_t. The expression - $(1 - i_t) s_{t-1}$, here $(1 - i_t)$ acts as a forget gate to control how much of the previous state's info is to be passed on to the next. On the other hand, LSTM equations are

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$$s_T = f_T \cdot s_{T-1} + i_T \cdot \tilde{s}_T$$

$$\tilde{s}_T = \sigma(W h_{T-1} + U x_T + b)$$

$$h_T = O_T \cdot s_T$$

$$i_T = \sigma(W_i h_{T-1} + U_i x_T + b_i)$$

$$O_T = \sigma(W_o h_{T-1} + U_o h_T + b_o)$$

$$f_T = \sigma(W_f h_{T-1} + U_f h_T + b_f)$$

Thus, three gates are used here with 2 states (1 for long term memory & one for current state output (h))

This increases the number of parameters as opposed to ~~the~~ GRU which uses single state and uses it itself as a masked forget gate. This makes ~~it~~ the GRU ~~a~~ computationally more efficient.

This enables GRU to be used in less resource devices as well like ~~auto complete~~ ~~translating~~ audio translation on a mobile device ~~to~~ itself.

GRU strikes balance ~~between~~ solving ~~to~~ the disadvantage of RNN (short memory) & computational complexity problem of LSTM (fewer parameters).



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	2b)	1. Autoencoder uses a hidden layer to encode the input into a compressable format & a decoder to decode the encoded input and decrease the original & derived inputs loss. $\hat{i} = f(w^*h + b)$ $h = g(w_i + b)$ 2. An the autocomplete encoder behaves similar to PCA if, a) A linear function is used at both decoder & encoder side b) MSE method is used to calculate & minimize error 3. The dimensions of hidden layer in undercomplete auto encoder is less than that of the input layer. 4. Hence compression of input takes place, maintaining all necessary info & variance of input data. 5. PCA is also used for dimensionality reduction (reducing the number of features) by choosing important ones. 6. Hence they are doing the similar task of input compression 7. Therefore, linear autocomplete encoder trained with MSE is learns subspace equivalent to PCA. 8. Sometimes autoencoders tend to learn the identity function i.e. they memorize the training data & don't generalize well on new data.

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9. Regularization techniques like L2 where a weight decay method is followed to avoid errors help with generalization.
10. Without regularization, weights remain same & don't generalize well.



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- 2a) 1. Modern usecases involve complex scenarios such as image, audio processing, etc.
2. ~~Audio~~ For example, an audio signal ~~may~~ involves multiple layers across the depth such as bass, ambient noise, ~~≠~~ pitch, etc.
3. A single filter with the same depth as audio signal can't ~~be~~ prove useful here as it will generate only single feature map meaning only partial information is captured.
4. We will need multiple filters with different depths to capture different patterns in the data.
5. This generates multiple feature maps capturing various information aspects.
6. A pooling function is applied to perform operations on this data.
7. Hence the parameters should be trained accordingly.

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1a) $\theta = [w, b]$... weight & bias

$$w_{t+1} = w_t - \eta \nabla w_t$$

$$b_{t+1} = b_t - \eta \nabla b_t$$

$$\nabla w = \frac{\partial}{\partial w} \left(\frac{1}{2} (xw - y)^2 \right)$$

$$= \frac{1}{2} \times 2(xw - y) \times \frac{\partial}{\partial w} (xw)$$

$$= (xw - y) \times \frac{\partial}{\partial w} \left(\frac{1}{1 + e^{-(wx+b)}} \right) \quad \dots (1)$$

$$\frac{\partial}{\partial w} \left(\frac{1}{1 + e^{-(wx+b)}} \right) = \frac{-1}{(1 + e^{-(wx+b)})^2} \times e^{-(wx+b)} \times -x$$

Take xw as $f(x)$

$\therefore$ From (1)

$$\frac{\partial}{\partial w} \left(\frac{1}{1 + e^{-(wx+b)}} \right) = f(x) (1 - f(x)) x$$

$\therefore$ From (1)

$$\nabla w = f(x - y) \times f(x) \times (1 - f(x)) \times x$$

Similarly

$$\nabla b = f(x - y) \times f(x) \times (1 - f(x))$$

$$\therefore w_{t+1} = w_t - \eta \times f(x - y) \times f(x) \times (1 - f(x)) \times x$$

$$\therefore w_{t+1} - w_t < 0 \quad (\text{we need less loss})$$

$\therefore$

$$\Delta w =$$

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- 3a) 1. LSTM makes use of forget, input & output gates to manage passing or rejection of information.
2. Due to this two states are maintained one for short term & second for long.
3. Hence vanishing gradient problem is solved here as prior information is maintained via rejection/passing of information to next state.

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Sr. No. 8500



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$$S_t = \sigma(W_{t-1} S_{t-1} + U X_t + b)$$

$$0 = \sigma(W_{t-1} S_{t-1} + b)$$

$$S_t \rightarrow \textcircled{2}$$

$$\uparrow$$

$$W$$